General relativity's predictions were poorly understood. Nobody realized that a sufficiently compact object *must* implode, and that the implosion will produce a black hole. Rather, Schwarzschild singularities (black holes) were imagined, incorrectly, to be objects that are hovering at or just inside their critical circumference, supported against gravity by some sort of internal force; Einstein therefore thought he could debunk black holes by showing that nothing supported by internal forces can be as small as the critical circumference.

If Einstein had suspected that "Schwarzschild singularities" can really exist, he might well have realized that implosion is the key to forming them and internal forces are irrelevant. But he was so firmly convinced they cannot exist (they "smelled wrong"; terribly wrong) that he had an impenetrable mental block against the truth—as did nearly all his colleagues.

In T. H. White's epic novel *The Once and Future King* there is a society of ants which has the motto, "Everything not forbidden is compulsory." That is *not* how the laws of physics and the real Universe work. Many of the things permitted by the laws of physics are so highly improbable that in practice they never happen. A simple and time-worn example is the spontaneous reassembly of a whole egg from fragments splattered on the floor: Take a motion picture of an egg as it falls to the floor and splatters into fragments and goo. Then run the motion picture backward, and watch the egg spontaneously regenerate itself and fly up into the air. The laws of physics permit just such a regeneration with time going forward, but it never happens in practice because it is highly improbable.

Physicists' studies of black holes during the 1920s and 1930s, and even on into the 1940s and 1950s, dealt only with the issue of whether the laws of physics *permit* such objects to exist—and the answer was equivocal: At first sight, black holes seemed to be permitted; then Einstein, Eddington, and others gave (incorrect) arguments that they are forbidden. In the 1950s, when those arguments were ultimately disproved, many physicists turned to arguing that black holes might be permitted by the laws of physics, but are so highly improbable that (like the reassembling egg) they never occur in practice.